Minchan Kim

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EDUCATION

Qingdao Daewon School

Senior | Expected 2027 | GPA: 3.984 / 4.0 (Unweighted)

Major focus: Aviation, Aeronautics/Aerospace Science and Technology

ENGINEERING & EXPERIENCE

Founder & President | COSMOS (Aero-Engineering Club)

Sep 2025 - Present | Qingdao Daewon School

Founded and led a 10-member aerospace engineering club through full theory to design to fabrication to test-validation project cycles.

Directed the Solid Propellant Rocket Thrust Test Stand project: two static-fire test iterations, diagnosing and resolving a data-logging failure between tests, then deriving peak thrust (190N), total impulse (270.6 N·s), and specific impulse (95.2s) from the results.

Founder & Developer | Tern

Mar 2026 - Present | Gyeonggi, South Korea

Built Tern (flytern.site), a flight booking web app using Next.js, the Duffel API, and React Three Fiber.

Designed and implemented "Civil Twilight," a custom glassmorphism design system.

Deputy Head of Technology | DAEMUN III (Daewon Model UN)

Aug 2026 - Present | Qingdao Daewon School

Serving as Deputy Head of Technology for DAEMUN III, contributing to the conference website's Next.js/Hono/PostgreSQL monorepo alongside the project lead and technology team.

Built the admin Teams screen and a per-committee 'publish approved resolutions' workflow, replacing an all-or-nothing publish step.

Vice President | Mechanics Club

Mar 2026 - Present | Qingdao Daewon School

Designed and built a 4-bit Ripple Carry Adder from scratch - researched binary arithmetic, logic gates (AND/OR/XOR), and half-/full-adder theory, then implemented and verified the circuit on a breadboard with logic gate ICs and LED indicators.

Conceived the initial concept for a flight-booking web app within the club, which I later took further and built independently as Tern (see the Tern project).

Aerospace & Space Camp Participant | Korea Aerospace University

Aug 2026 - Aug 2026 | Goyang, South Korea

Participated in an intensive aerospace program covering flight simulation (FTD), air traffic control (ATC), cube satellite design, and autonomous drone systems.

Visited Korean Air and Asiana Airlines headquarters, touring aircraft hangars and completing evacuation slide and safety training.

Workshop Participant | Shandong University

Jul 2025 - Aug 2025 | Weihai, Shandong, China

Participated in a cultural immersion workshop focused on Chinese language and culture.

Engaged in daily Mandarin practice sessions and cultural activities alongside an international cohort, including students from Russia.

TECHNICAL SKILLS

FreeCAD, OpenSCAD, Arduino, System Architecture, Next.js, Typescript, React Three Fiber, GSAP, APIs, PostgreSQL, Digital Logic, React, Python

RESEARCH & SELECTED PROJECTS

F-22-Inspired RC Aircraft: Aerodynamic Design & Flight Stability Research

Designed, built, and flight-tested an F-22-inspired RC aircraft, then wrote a research paper on the correlation between its aerodynamic design variables and flight stability - 1st place, Physics Division, school research paper competition.

Automated CIWS Drone Tracking Turret

Engineered an automated Close-In Weapon System (CIWS) style turret that autonomously detects and tracks moving drone targets.

Project AEROVIEW - Open-Circuit Suction Wind Tunnel

Directing the parametric CAD design and fabrication of an open-circuit suction wind tunnel to compare airflow over multiple NACA airfoil profiles - in progress, continuing next semester.

HONORS & CERTIFICATIONS

Aerospace Work Experience - Airbus | Springpod | March 2026

Claude Development & AI Fluency | Anthropic | July 2026

SAT | The College Board | March 2026

TOEFL iBT | ETS | July 2026

1st Place, School Science Fair | Qingdao Daewon School | 2025 - 2026

1st Place (Physics Division), School Research Paper Competition - "Analysis of the Correlation Between Aerodynamic Design Variables and Flight Stability of an F-22-Based RC Aircraft" | Qingdao Daewon School | G11, 2025 - 2026

2nd Place, School-wide Writing Contest - Top Excellence Award, prose essay on "What I'd Want to Tell My 8-Year-Old Self" | Qingdao Daewon School | G10, 2024

COMMUNITY & LEADERSHIP

Qingdao Tonghang House | Volunteer (Intellectual Disability Welfare Facility)

October 2025 - November 2025

Facilitated daily structured activities for residents and provided operational support to facility staff for ongoing programs.

Student Council | Planning Department Vice President

2025 - March 2026

Organized joint domestic/international student events as part of the Planning Department, coordinating cross-department logistics and program design.

PSYCHOFISH | Member

2024 - 2025

Ran an anonymous peer-counseling channel over KakaoTalk open chat, listening to classmates' concerns and offering thoughtful responses.

Point of View | Member, PR Team

2024

Designed rules and posters for a school-wide Manito (secret friend) event as part of the PR team.

Support Group | Group Leader

2024 - 2026 (4 semesters through G11)

Participated in the school's semesterly Support Group program all four times offered through 11th grade, a small-group peer mentoring initiative run each semester.

Goyang Mayor's Cup Kkumnamu Three-Cushion Billiards Tournament | Participant

July 2026

A personal hobby since middle school, mostly playing four-ball carom - competed in this three-cushion tournament hosted by the Goyang City Billiards Federation, stepping outside my usual game into a more geometrically demanding discipline.